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CUET UG

Chemistry

Coordination Compounds

Official Syllabus Unit V · Class 12 · Comprehensive Theory, Bonding Mechanisms & 15 Exam-Pattern MCQs

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

Sub-Topic	Official Syllabus Parameters (NTA CUET UG Chemistry)	Core Exam Competencies Tested
Definitions & Werner's Theory	General introduction to coordination compounds, Werner's theory, double salts vs complexes, coordination entity, central atom/ion, ligands, coordination number, coordination sphere, oxidation number, homoleptic & heteroleptic.	Identifying primary vs secondary valencies, calculating coordination and oxidation numbers, predicting precipitation moles with $AgNO_3$ and molar conductance.
IUPAC Nomenclature	IUPAC nomenclature of mononuclear coordination compounds.	Applying systematic naming rules: cation before anion, alphabetical ligand order, anionic -o suffixes, anionic metal -ate suffixes, Roman oxidation numbers.
Isomerism	Structural isomerism (ionisation, solvate/hydrate, linkage, coordination) and stereoisomerism (geometrical cis/trans, fac/mer, and optical chirality).	Classifying isomer pairs, predicting stereoisomers in $[Ma_2b_2]$, $[Ma_4b_2]$, $[Ma_3b_3]$, and determining optical activity in bis/tris-chelate complexes.
Bonding: VBT & CFT	Valence Bond Theory (hybridisation, inner/outer orbital, magnetic moment, limitations), Crystal Field Theory (octahedral and tetrahedral splitting, spectrochemical series, Δ_o vs P, colour).	Predicting hybridisations ($sp^3, dsp^2, sp^3d^2, d^2sp^3$), evaluating high-spin vs low-spin states, calculating $\mu_s = \sqrt{n(n+2)}$ BM, explaining colour origins.
Metal Carbonyls & Applications	Nature of bonding in metal carbonyls (synergic σ-donation and π-back-bonding), importance and applications of coordination compounds in metallurgy, analysis, and biology.	Explaining $M-C$ bond strengthening, identifying coordination catalysts, recognizing biological metal centres (chlorophyll, haemoglobin, vitamin B_{12}).

EXAM PATTERN

CUET UG Paper Pattern & Weightage: The Chemistry paper carries **50 compulsory questions** in **60 minutes** (+5 marks for correct, −1 mark for incorrect). Unit V (Coordination Compounds) contributes 4 to 6 questions focusing on Werner's precipitation

stoichiometry, IUPAC naming, *fac*/*mer* and optical isomerism, and CFT electronic configurations.

Module 1: Foundational Vocabulary & Werner's Theory

A **coordination compound** consists of a central metal atom or ion bonded to a group of surrounding ions or neutral molecules (**ligands**) via dative coordinate bonds, where both electrons in each bond are supplied by the ligand.

- **Complex vs Double Salt:**
  - **Double Salt:** Stable only in crystalline solid state; completely dissociates into constituent simple ions when dissolved in water (e.g. Mohr's salt: $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ gives individual tests for Fe^{2+}, NH_4^+, and SO_4^{2-}; Carnallite: $KCl \cdot MgCl_2 \cdot 6H_2O$).
  - **Coordination Complex:** Retains its structural identity in aqueous solution; the complex coordination entity does not dissociate into individual constituent ions (e.g. $K_4[Fe(CN)_6]$ dissociates into $4K^+$ and the stable complex anion $[Fe(CN)_6]^{4-}$, yielding zero free Fe^{2+} or CN^- ions).

1. Anatomy of a Coordination Complex

Taking potassium hexacyanidoferrate(II), $K_4[Fe(CN)_6]$, as the canonical archetype:

- **Coordination Entity:** The enclosed square-bracketed unit, $[Fe(CN)_6]^{4-}$, comprising the central metal and bonded ligands acting as a unified charged or neutral species.
- **Central Atom / Ion:** The transition metal (Fe^{2+}) acting as a Lewis acid by accepting electron pairs into its vacant $(n-1)d, ns$, and np orbitals.
- **Ligands:** The Lewis base molecules or ions donating electron lone pairs to the metal center.
- **Coordination Number (CN):** The total number of ligand donor atoms directly bonded to the central metal. In $[Fe(CN)_6]^{4-}$, CN = 6. For bidentate ligands (like oxalate or ethylenediamine), each ligand contributes 2 donor atoms (e.g. in $[Co(en)_3]^{3+}$, CN $= 3 \times 2 = 6$).
- **Coordination Sphere:** Everything enclosed inside the square brackets []. Ions outside are **counter ions** which dissociate in water.
- **Oxidation Number:** The residual charge on the central atom when all ligands are removed with their donor pairs: for $[Fe(CN)_6]^{4-}$, $x + 6(-1) = -4 \rightarrow x = +2$.

2. Classification of Ligands by Denticity

1. **Monodentate Ligands:** Coordinate via a single donor atom (e.g. $Cl^-, Br^-, H_2O, NH_3, CN^-$).
2. **Bidentate Ligands:** Possess two donor atoms capable of simultaneous binding:
   - Ethane-1,2-diamine (en, neutral): $H_2N-CH_2-CH_2-NH_2$ (two N donor atoms).
   - Oxalate ion (ox^{2-} or $C_2O_4^{2-}$, anionic): $^-OOC-COO^-$ (two O donor atoms).

3. **Polydentate Ligands:** Bind through multiple donor atoms. E.g. Ethylenediaminetetraacetate ($EDTA^{4-}$) is a **hexadentate ligand** binding via two nitrogen atoms and four carboxylate oxygen atoms.
4. **Chelate Ligands & The Chelate Effect:** Di- or polydentate ligands that coordinate to a single central metal form 5- or 6-membered heterocyclic rings termed **chelate rings**. Complexes containing chelate rings (e.g. $[Fe(ox)_3]^{3-}$) exhibit dramatically higher thermodynamic stability than analogous complexes containing only monodentate ligands. This enhanced stability is termed the **chelate effect** (driven by a favorable positive entropy change, $\Delta S > 0$).
5. **Ambidentate Ligands:** Monodentate ligands possessing two distinct potential donor atoms, but coordinating through only one atom at a time:
   - Nitrite ion: Coordinates via nitrogen (NO_2^-, nitro or nitrito-κN) OR via oxygen (ONO^-, nitrito or nitrito-κO).
   - Thiocyanate ion: Coordinates via sulphur (SCN^-, thiocyanato) OR via nitrogen (NCS^-, isothiocyanato).

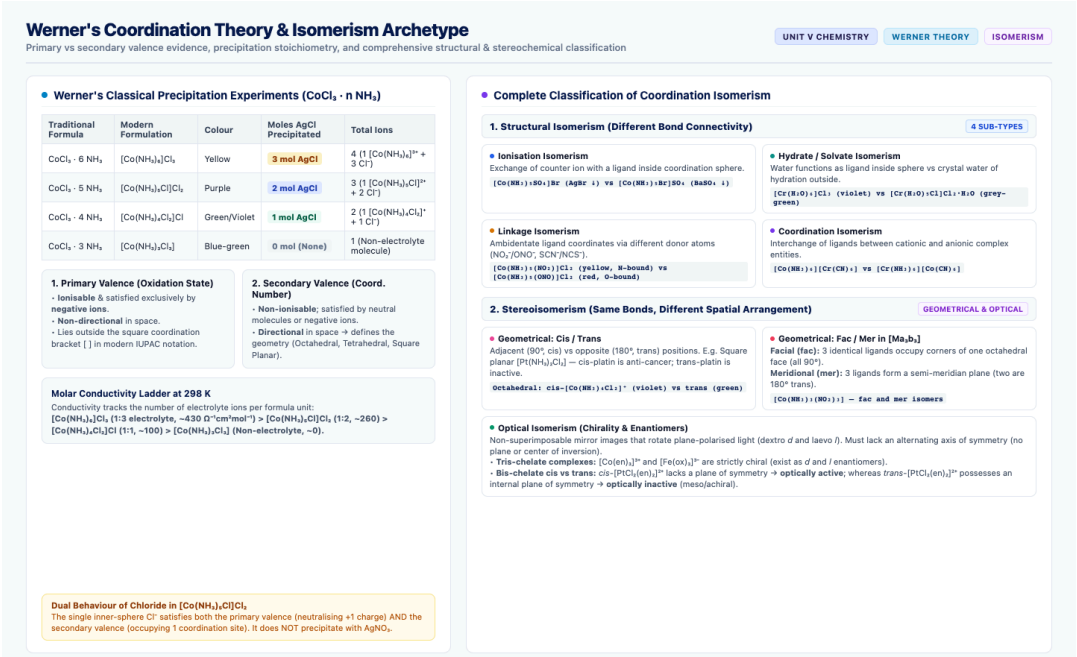


Figure 1: Werner's classical precipitation experiments on cobalt ammines, dual valency principles, and systematic classification of structural and stereoisomerism in coordination compounds.

Quantitative Analysis: Werner's Coordination Theory

Formulated by Alfred Werner in 1893, this theory resolved the riddle of how metals with fixed classical valencies could bind additional neutral molecules.

1. Postulates of Werner's Theory

1. **Primary Valence (Oxidation State):** Ionisable; satisfied exclusively by negative ions; non-directional in space.
2. **Secondary Valence (Coordination Number):** Non-ionisable; satisfied by neutral molecules or negative ions; possesses fixed spatial orientations (directional), determining the stereochemical geometry of the complex.

2. Classical Experimental Evidence (Precipitation & Molar Conductivity)

Werner treated series of cobalt(III) chloride-ammonia adducts with excess aqueous silver nitrate ($AgNO_3$) at 298 K, measuring moles of AgCl precipitated per mole of complex:

- **$CoCl_3 \cdot 6NH_3 \rightarrow [Co(NH_3)_6]Cl_3$ (Yellow):** Precipitates **3 moles of AgCl**. All 3 chlorides are in the outer ionisation sphere. Molar conductance $\Lambda_m \approx 430$ Omega^{-1}cm^{2}mol^{-1} (1:3 electrolyte, 4 total ions).
- **$CoCl_3 \cdot 5NH_3 \rightarrow [Co(NH_3)_5\ Cl]Cl_2$ (Purple):** Precipitates **2 moles of AgCl**. One chloride has moved inside the coordination sphere to satisfy a secondary valence. Conductance ≈ 260 (1:2 electrolyte, 3 total ions).
- **$CoCl_3 \cdot 4NH_3 \rightarrow [Co(NH_3)_4Cl_2]\ Cl$ (Green/Violet):** Precipitates **1 mole of AgCl**. Two chlorides are inside the coordination sphere. Conductance ≈ 100 (1:1 electrolyte, 2 total ions).
- **$CoCl_3 \cdot 3NH_3 \rightarrow [Co(NH_3)_3Cl_3]$ (Blue-green):** Precipitates **0 moles of AgCl**. All 3 chlorides satisfy secondary valencies inside the sphere. Non-electrolyte ($\Lambda_m \approx 0$).

Module 2: IUPAC Nomenclature of Coordination Entities

1. Systematic Nomenclature Rules

1. **Order of Naming:** The cation is named first, followed by the anion, regardless of whether the coordination entity is the cation or the anion (e.g. in $K_4[Fe(CN)_6]$, potassium is named first; in $[Co(NH_3)_6]Cl_3$, the complex cation is named first).
2. **Naming Within the Coordination Sphere:** Ligands are listed in strict **alphabetical order** of their names, preceding the central metal name. Multiplicative prefixes (di, tri, tetra, bis, tris) do NOT affect alphabetical order.
3. **Ligand Name Terminations:**
   - **Anionic Ligands:** End in **-o**. Examples: Cl^- (chlorido), CN^- (cyanido), OH^- (hydroxido), NO_2^- (nitrito-κN), ONO^- (nitrito-κO), $C_2O_4^{2-}$ (oxalato), SO_4^{2-} (sulfato).
   - **Neutral Ligands:** Retain common names, with four key exceptions: H_2O (**aqua**), NH_3 (**ammine** with double 'm'), CO (**carbonyl**), and NO (**nitrosyl**).
4. **Multiplying Prefixes:** Simple prefixes (di, tri, tetra, penta, hexa) designate ligand numbers. When a ligand's name already contains a numeral prefix (e.g. ethylenediamine), multiplicative prefixes **bis-**, **tris-**, **tetrakis-** are used, and the ligand name is placed in parentheses.
5. **Metal Name Suffixes:**
   - In **cationic** or **neutral** complexes: The central metal retains its regular English elemental name (e.g. cobalt, chromium, iron, platinum).
   - In **anionic** complexes: The metal name must terminate in **-ate** (e.g. cobaltate, chromate, nickelate). For metals with classical Latin roots: iron → **ferrate**, copper → **cuprate**, silver → **argentate**, lead → **plumbate**, tin → **stannate**, gold → **aurate**.
6. **Oxidation State:** Indicated by a Roman numeral in parentheses immediately following the metal name without any intervening space: e.g. iron(II), cobalt(III).

Formula	Official IUPAC Name
$\left[Co(NH_3)_6\right]Cl_3$	hexaamminecobalt(III) chloride
$\left[Co(NH_3)_5\ Cl\right]Cl_2$	pentaamminechloridocobalt(III) chloride
$K_3[Fe(CN)_6]$	potassium hexacyanidoferrate(III)
$K_4[Fe(CN)_6]$	potassium hexacyanidoferrate(II)
$\left[Pt(NH_3)_2Cl_2\right]$	diamminedichloridoplatinum(II)
$[Cr(en)_3]Cl_3$	tris(ethane-1,2-diamine)chromium(III) chloride
$\left[Co(NH_3)_4(H_2\ O)\ Cl\right]Cl_2$	tetraammineaquachloridocobalt(III) chloride
$[Ni(CO)_4]$	tetracarbonylnickel(0)
$K_2[Zn(OH)_4]$	potassium tetrahydroxidozincate(II)

Module 3: Structural & Stereoisomerism

Isomers are compounds that possess identical molecular formulas but differ in their chemical structures or 3D spatial arrangements.

1. Structural Isomerism (Different Atom Connectivity)

1. **Ionisation Isomerism:** Occurs when a counter-ion exchanges positions with a ligand inside the coordination sphere, yielding different ions in aqueous solution:

$$\left[Co(NH_3)_5(SO_4)\right]\ Br\ \ (\text{Red; gives yellow}\ \ AgBr\downarrow\ \text{with}\ \ AgNO_3)$$

$$\left[Co(NH_3)_5\ Br\right]SO_4\ \ (\text{Red-violet; gives white}\ \ BaSO_4\downarrow\ \text{with}\ \ BaCl_2)$$

2. **Hydrate / Solvate Isomerism:** Water acts either as a direct coordinate ligand inside the sphere or as crystal lattice water of hydration outside:
   - $\left[Cr(H_2\ O)_6\right]Cl_3$ (Violet; precipitates 3 mol AgCl).
   - $\left[Cr(H_2\ O)_5\ Cl\right]Cl_2\cdot H_2\ O$ (Grey-green; precipitates 2 mol AgCl).
   - $\left[Cr(H_2\ O)_4Cl_2\right]\ Cl\cdot 2H_2\ O$ (Dark green; precipitates 1 mol AgCl).
3. **Linkage Isomerism:** Arises when an ambidentate ligand binds through different donor atoms:
   - Yellow $\left[Co(NH_3)_5(NO_2)\right]Cl_2$ (nitro isomer; *N*-coordinated).
   - Red $\left[Co(NH_3)_5(ONO)\right]Cl_2$ (nitrito isomer; *O*-coordinated).
4. **Coordination Isomerism:** Occurs in salts containing both cationic and anionic complex ions, through interchange of ligands between the two metal centers:

$$\left[Co(NH_3)_6\right]\left[Cr(CN)_6\right]\quad\text{versus}\quad\left[Cr(NH_3)_6\right]\left[Co(CN)_6\right]$$

2. Stereoisomerism (Same Bonds, Different Spatial Arrangement)

1. **Geometrical Isomerism (Cis/Trans & Fac/Mer):**
   - **Square Planar $[Ma_2b_2]$:** *cis*-$\left[Pt(NH_3)_2Cl_2\right]$ (cis-platin, anti-cancer drug, chlorides adjacent at 90°) versus *trans*-$\left[Pt(NH_3)_2Cl_2\right]$ (chlorides opposite at 180°, therapeutically inactive). Note: Tetrahedral complexes cannot show geometrical isomerism because all four ligand positions are adjacent (109.5°).

- **Octahedral $[Ma_4b_2]$:** *cis*-$\left[Co(NH_3)_4Cl_2\right]^+$ (violet, chlorides at 90°) versus *trans*-$\left[Co(NH_3)_4Cl_2\right]^+$ (green, chlorides at 180°).
- **Octahedral $[Ma_3b_3]$ (Facial / Meridional):**
  - **Facial (fac):** Three identical ligands occupy the corners of one octahedral face (all mutually cis, 90°). E.g. *fac*-$\left[Co(NH_3)_3(NO_2)_3\right]$.
  - **Meridional (mer):** Three identical ligands span across a semi-meridian plane (two are trans at 180°, one is cis at 90°). E.g. *mer*-$\left[Co(NH_3)_3(NO_2)_3\right]$.

2. **Optical Isomerism (Chirality & Enantiomers):**
   - Optical isomers are non-superimposable mirror images (**enantiomers**) that rotate plane-polarized light in opposite directions (dextrorotatory d and laevorotatory l).
   - **Requirement:** The complex must lack an alternating axis of symmetry (no center of inversion or plane of symmetry).
   - **Tris-Chelate Complexes:** $[Co(en)_3]^{3+}$ and $[Fe(ox)_3]^{3-}$ are strictly chiral and exist as separable d and l enantiomers.
   - **Bis-Chelate Octahedral Complexes $[Ma_2(b-b)_2]$:**
     - *cis*-$[Pt\ Cl_2(en)_2]^{2+}$ lacks a plane of symmetry → **Optically active** (chiral).
     - *trans*-$[Pt\ Cl_2(en)_2]^{2+}$ possesses an internal plane of symmetry passing through Pt and both Cl atoms → **Optically inactive** (achiral/meso).

Module 4: Valence Bond Theory (VBT)

Proposed by Linus Pauling, VBT describes coordination bonding as the overlap of vacant hybridised metal orbitals with lone-pair donor orbitals of ligands.

1. Hybridisation Schemes & Geometries

CN	Hybridisation	Geometry	Representative Complex Archetype
4	sp^3	Tetrahedral	$[NiCl_4]^{2-}$ (Paramagnetic, $\mu = 2.83$ BM)
4	dsp^2	Square Planar	$[Ni(CN)_4]^{2-}$ (Diamagnetic, $\mu = 0$ BM)
6	d^2sp^3	Inner Octahedral	$\left[Co(NH_3)_6\right]^{3+}$, $[Fe(CN)_6]^{4-}$ (Low-spin)
6	sp^3d^2	Outer Octahedral	$[CoF_6]^{3-}$, $\left[Fe(H_2O)_6\right]^{2+}$ (High-spin)

2. Inner vs Outer Orbital Complexes

- **Inner Orbital Complex (d^2sp^3):** In the presence of strong-field ligands (e.g. CN^-, CO), unpaired electrons in the $(n-1)d$ subshell are forced to pair up, leaving two vacant $3d$ orbitals available for d^2sp^3 hybridisation. These complexes have low spin (diamagnetic or low paramagnetism).
- **Outer Orbital Complex (sp^3d^2):** In the presence of weak-field ligands (e.g. F^-, Cl^-, water), the ligand field is insufficient to force electron pairing. Hybridisation uses vacant

outer $4d$ orbitals (sp^3d^2), leaving maximum unpaired electrons. These are high-spin complexes.

- **Limitations of VBT:** Cannot explain why ligands differ in field strength, cannot predict electronic absorption spectra (colours), and treats magnetic moment data as an empirical assumption rather than a derived property.

3. Step-by-Step VBT Electronic Configuration Analysis

To predict the hybridisation, magnetic character, and geometry of any coordination complex in CUET UG:

1. **Identify Oxidation State of Central Metal:** Calculate the oxidation state x from ligand and overall charges.
2. **Write Outer Electronic Configuration:** Write the ground-state valence configuration (e.g. $Fe^{2+} = 3d^6, Fe^{3+} = 3d^5, Co^{3+} = 3d^6, Ni^{2+} = 3d^8$).
3. **Evaluate Ligand Field Strength:**
   - **Strong Ligands** (CN^-, CO, en, and NH_3 with +3 cations): Force pairing of $(n-1)d$ electrons where possible.
   - **Weak Ligands** ($F^-, Cl^-, Br^-, I^-, H_2O$): Cannot overcome pairing energy P; electrons remain unpaired according to Hund's rule.
4. **Assign Orbitals for Coordinate Dative Pairs:**
   - If two internal $3d$ orbitals become vacant: d^2sp^3 (**Inner orbital octahedral, low-spin**).
   - If internal $3d$ orbitals remain unavailable: sp^3d^2 (**Outer orbital octahedral, high-spin**).

Complex Ion	Metal Config	Ligand Field	Hybridisation	Unpaired (n)	Magnetic Nature & μ_s
$[Fe(CN)_6]^{4-}$	$3d^6$	Strong (CN^-)	d^2sp^3 (Inner)	0	Diamagnetic ($\mu = 0$)
$\left[Fe(H_2O)_6\right]^{2+}$	$3d^6$	Weak (H_2O)	sp^3d^2 (Outer)	4	Paramagnetic ($\mu = 4.90$ BM)
$[Fe(CN)_6]^{3-}$	$3d^5$	Strong (CN^-)	d^2sp^3 (Inner)	1	Paramagnetic ($\mu = 1.73$ BM)
$\left[Co(NH_3)_6\right]^{3+}$	$3d^6$	Strong (NH_3 on +3)	d^2sp^3 (Inner)	0	Diamagnetic ($\mu = 0$)
$[CoF_6]^{3-}$	$3d^6$	Weak (F^-)	sp^3d^2 (Outer)	4	Paramagnetic ($\mu = 4.90$ BM)
$[Ni(CN)_4]^{2-}$	$3d^8$	Strong (CN^-)	dsp^2 (Sq. Planar)	0	Diamagnetic ($\mu = 0$)
$[NiCl_4]^{2-}$	$3d^8$	Weak (Cl^-)	sp^3 (Tetrahedral)	2	Paramagnetic ($\mu = 2.83$ BM)
$[Ni(CO)_4]$	$3d^{10}$ ($s \to d$)	Strong (CO)	sp^3 (Tetrahedral)	0	Diamagnetic ($\mu = 0$)

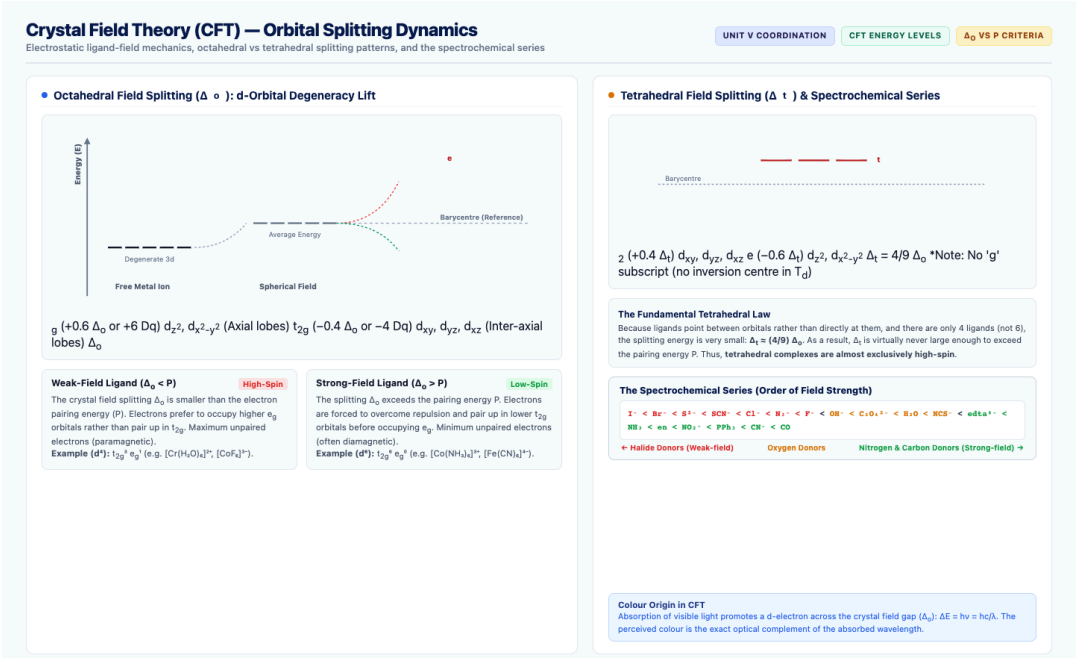


Figure 2: Crystal Field Theory orbital splitting in octahedral (Δ_o) and tetrahedral (Δ_t) fields, high-spin vs low-spin decision criteria, and the complete Spectrochemical Series.

Module 5: Crystal Field Theory (CFT) & Metal Carbonyls

Developed by Hans Bethe and J.H. Van Vleck, CFT considers the metal-ligand bond as purely electrostatic (ionic point charges for anions or point dipoles for neutral molecules).

1. Octahedral Crystal Field Splitting (Δ_o)

In an isolated gaseous metal atom, all five d-orbitals are degenerate.

- In an octahedral complex, six ligands approach the metal along the three Cartesian axes (x, y, z).
- **The e_g Set ($d_{x^2-y^2}, d_{z^2}$):** Orbital lobes point directly along axes toward approaching ligands. They experience maximal electrostatic repulsion and are raised in energy by $+0.6\Delta_o$ (+6 Dq) above the barycentre.
- **The t_{2g} Set (d_{xy}, d_{yz}, d_{xz}):** Orbital lobes point between Cartesian axes. They experience less repulsion and are stabilized by $-0.4\Delta_o$ (−4 Dq) below the barycentre.
- The energy separation between the two sets is the **octahedral crystal field splitting energy (Δ_o)**.

2. High-Spin vs Low-Spin Decision Criteria (Δ_o vs P)

For d^1, d^2, d^3 configurations, electrons enter t_{2g} singly obeying Hund's rule: $t_{2g}^1, t_{2g}^2, t_{2g}^3$. At d^4, two possibilities emerge, governed by the balance between Δ_o and the **Pairing Energy (P)**:

1. **Weak-Field Ligand ($\Delta_o < P$):** Energy required to promote an electron across Δ_o is less than the energy required to pair electrons. The 4th electron enters the higher e_g orbital:

$$t_{2g}^3 e_g^1 \quad \text{High-Spin Complex (Maximum Unpaired Electrons)}$$

   Example: $[CoF_6]^{3-}$ (d^6) has $\Delta_o < P \rightarrow t_{2g}^4 e_g^2$ (4 unpaired electrons, paramagnetic).
2. **Strong-Field Ligand ($\Delta_o > P$):** Splitting Δ_o is very large. Electrons pair up in the lower t_{2g} set before occupying e_g:

$t_{2g}^4 e_g^0$ Low-Spin Complex (Minimum Unpaired Electrons)

Example: $\left[\mathrm{Co(NH_3)_6}\right]^{3+}$ (d^6) has $\Delta_o > P \rightarrow t_{2g}^6 e_g^0$ (0 unpaired electrons, diamagnetic).

3. Tetrahedral Crystal Field Splitting (Δ_t)

In tetrahedral geometry, four ligands approach between the axes:

- The splitting is inverted: the e set ($d_{x^2-y^2}, d_{z^2}$) is lower, while the t_2 set (d_{xy}, d_{yz}, d_{xz}) is higher.
- **Subscript Rule:** The subscript 'g' (gerade) is omitted because a tetrahedron lacks an inversion centre.
- **Small Splitting Energy:** $\Delta_t = \frac{4}{9}\Delta_o$. Because Δ_t is always substantially smaller than P, **tetrahedral complexes are virtually always high-spin**.

4. The Spectrochemical Series

An experimentally determined order of ligand field strength based on absorption spectra:

$$^- < S^{2-} < \mathrm{SCN^-} < \mathrm{Cl^-} < N_3^- < F^- < \mathrm{OH^-} < C_2O_4^{2-} < \mathrm{H_2\ O} < \mathrm{NCS^-} < \mathrm{EDTA^{4-}} < \mathrm{NH_3} < \mathrm{en} < \mathrm{NO_2^-} < \mathrm{PPh_3} < \mathrm{CN}$$

- **Halide donors** ($I^- \rightarrow F^-$): Weakest field ligands (high-spin).
- **Oxygen donors** ($\mathrm{OH^-}, \mathrm{ox^{2-}}, \mathrm{H_2\ O}$): Intermediate field ligands.
- **Nitrogen & Carbon donors** ($\mathrm{NH_3}, \mathrm{en}, \mathrm{CN^-}, \mathrm{CO}$): Strongest field ligands (low-spin).

5. Synergic Bonding in Metal Carbonyls

In homoleptic metal carbonyls (e.g. $[\mathrm{Ni(CO)_4}]$, $[\mathrm{Fe(CO)_5}]$, $[\mathrm{Cr(CO)_6}]$), bonding comprises two mutually reinforcing components (**synergic effect**):

1. **σ-Dative Bond:** Ligand CO donates a lone pair from its carbon atom into a vacant metal d-orbital ($C \rightarrow M$).
2. **π-Back-Bond:** The filled metal d-orbital overlaps with the empty antibonding π^* molecular orbital of CO, transferring electron density back from metal to ligand ($M \rightarrow C$).

- This synergic loop increases $M-C$ bond order and strength while weakening the $C-O$ bond (lowering the $C-O$ stretching frequency in IR spectroscopy).

6. Crystal Field Stabilization Energy (CFSE) & Colour Origins

1. **Mathematical Formulation of CFSE:** In an octahedral complex, the net energetic stabilization relative to the hypothetical spherical field is:

$$\mathrm{CFSE}_o = \left[-0.4n\left(t_{2g}\right) + 0.6n\left(e_g\right)\right]\Delta_o + mP$$

where $n\left(t_{2g}\right)$ and $n\left(e_g\right)$ are electron counts in the respective orbital sets, and m is the number of extra electron pairs formed relative to the free gaseous ion.

2. **Origin of Absorption Spectra & Complementary Colour:**
   - Transition metal complexes owe their vibrant colours to **electronic $d-d$ transitions**.
   - When white light strikes a complex with partially filled d-subshell (d^1 to d^9), a t_{2g} electron absorbs a photon of energy matching Δ_o:

$$\Delta_o = h\nu = \frac{hc}{\lambda}$$

- The electron is promoted from the stabilized t_{2g} set to the higher e_g set. The complex transmits the remaining wavelengths, producing the perceived **complementary colour**.
- **Example:** Hexaaquatitanium(III) ion, $\left[\mathrm{Ti(H_2\ O)_6}\right]^{3+}$ ($3d^1$), absorbs yellow-green light ($\lambda \approx 498$ nm) and consequently appears brilliant **violet**.
- **Condition for Colour:** The metal must possess **partially filled *d*-orbitals** (d^1 to d^9).
  - d^0 complexes (e.g. Sc^{3+}, Ti^{4+}) have no *d*-electrons to promote → **Colourless**.
  - d^{10} complexes (e.g. $Cu^+(3d^{10}), Zn^{2+}(3d^{10})$) have completely filled e_g sets (no vacancy for excitation) → **Colourless**.
- **Special Exception (Charge Transfer Spectra):** Permanganate ($KMnO_4, Mn^{7+} = d^0$, intense purple) and Dichromate ($K_2Cr_2O_7, Cr^{6+} = d^0$, bright orange) are deeply coloured NOT by $d-d$ transitions, but by **Ligand-to-Metal Charge Transfer (LMCT)** from filled oxide *p*-orbitals to empty metal *d*-orbitals.

Module 6: Stability & Practical Applications

1. Stability of Coordination Compounds in Solution

The formation of a complex in solution proceeds via stepwise equilibrium constants ($K_1, K_2, ...$):

$$M + 4L \leftrightarrow ML_4; \quad \beta_4 = K_1K_2K_3K_4 = \frac{[ML_4]}{[M][L]^4}$$

- **Factors Elevating Stability:** High charge density on the central metal ion (small radius, high positive charge), strong-field ligands (high basicity), and the presence of multidentate chelating ligands (**chelate effect**).

2. High-Yield Industrial & Biological Applications

- **Metallurgy:** Extraction of Gold and Silver via cyanide leaching ($4\ Au + 8CN^- + 2H_2\ O + O_2 \rightarrow 4[Au(CN)_2]^- + 4OH^-$), followed by reduction with Zinc dust. Refining of Nickel by the Mond process via volatile $[Ni(CO)_4]$.
- **Analytical Chemistry:** Estimation of water hardness (Ca^{2+} and Mg^{2+}) by complexometric titration with $EDTA^{4-}$. Detection of Ni^{2+} using dimethylglyoxime (DMG), yielding a brilliant scarlet-red chelate precipitate $[Ni(DMG)_2]$.
- **Biological Systems:**
  - **Chlorophyll:** Coordination complex of **Magnesium (Mg^{2+})** with a porphyrin ring (photosynthesis).
  - **Haemoglobin:** Coordination complex of **Iron (Fe^{2+})** with porphyrin (oxygen carrier).
  - **Vitamin B_{12} (Cyanocobalamin):** Coordination complex of **Cobalt (Co^{3+})** (anti-pernicious anaemia).
- **Medicine: cis-Platin** (*cis*-$\left[\mathrm{Pt(NH_3)_2Cl_2}\right]$) is an indispensable anti-cancer chemotherapeutic agent that binds to and cross-links cancer cell DNA. Excess copper/iron poisoning is treated with chelating ligands like D-penicillamine or desferrioxamine B; lead poisoning is treated with Ca-$EDTA^{2-}$.

3. Stepwise vs Overall Stability Constants

The formation of complex ML_4 occurs in sequential steps, each governed by a stepwise stability constant K:

$$M + L \leftrightarrow ML \quad \left(K_1 = \frac{[ML]}{[M][L]}\right)$$

$$ML + L \leftrightarrow ML_2 \quad \left(K_2 = \frac{[ML_2]}{[ML][L]}\right)$$

$$ML_2 + L \leftrightarrow ML_3 \quad \left(K_3 = \frac{[ML_3]}{[ML_2][L]}\right)$$

$$ML_3 + L \leftrightarrow ML_4 \quad \left(K_4 = \frac{[ML_4]}{[ML_3][L]}\right)$$

The overall stability constant β_4 is the product of all stepwise constants:

$$\beta_4 = K_1 \cdot K_2 \cdot K_3 \cdot K_4 \quad \Longleftrightarrow \quad \log \beta_4 = \log K_1 + \log K_2 + \log K_3 + \log K_4$$

- **Fundamental Magnitude Trend:** $K_1 > K_2 > K_3 > K_4$.
- **Thermodynamic Justification:**
  1. **Statistical Factor:** As ligands bind, the number of remaining vacant coordination positions on the metal decreases, while the probability of ligand dissociation increases.
  2. **Steric Factor:** Increasing steric crowding around the coordination sphere impedes the approach of subsequent bulky ligands.
  3. **Electrostatic Factor:** Binding of anionic ligands diminishes the effective positive charge of the central metal cation.

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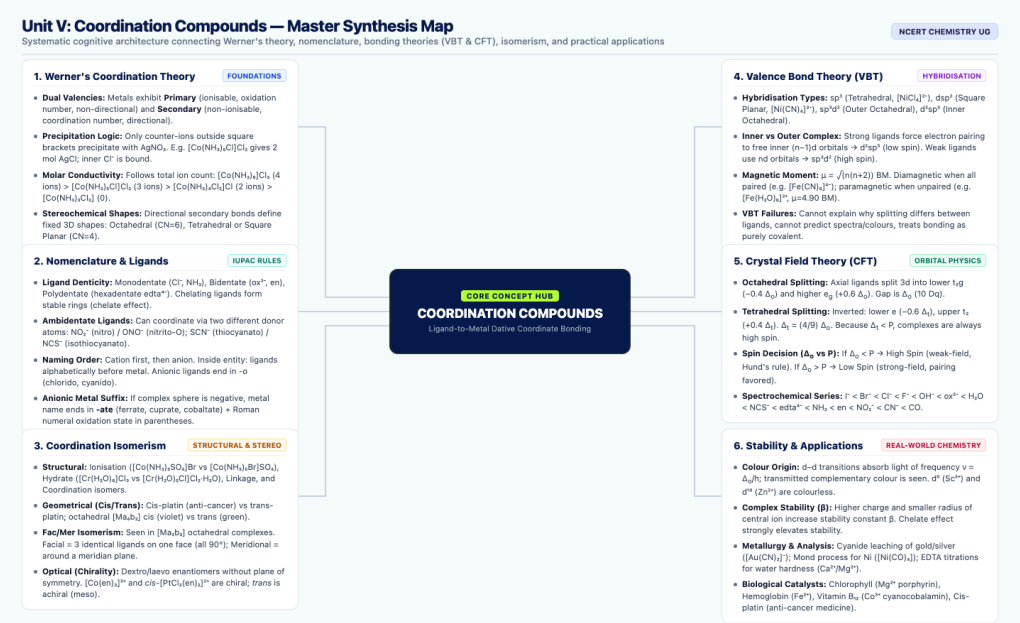


Figure 3: Master Synthesis Architecture: Complete Unit V cognitive roadmap integrating Werner's theory, nomenclature, bonding theories (VBT & CFT), isomerism, and practical applications.

UNIT V MASTER FORMULA & HIGH-YIELD CHEATSHEET

Core Concept / Law	Expression / Formulation	Application & CUET Problem Strategy
Werner Precipitation	Moles AgCl = Number of ionisable Cl^- outside []	Count outer counter-ions: $\left[Co(NH_3)_6\right]Cl_3 \rightarrow 3$ AgCl; $\left[Co(NH_3)_5Cl\right]Cl_2 \rightarrow 2$ AgCl; $\left[Co(NH_3)_4Cl_2\right]Cl \rightarrow 1$ AgCl.
Spin-Only Moment	$\mu_s = \sqrt{n(n+2)}$ BM	Compute unpaired electrons: $n=1 \rightarrow 1.73, n=2 \rightarrow 2.83, n=3 \rightarrow 3.87, n=4 \rightarrow 4.90, n=5 \rightarrow 5.92$ BM.
Octahedral Splitting	$e_g \rightarrow +0.6\Delta_o(+6$ Dq); $t_{2g} \rightarrow -0.4\Delta_o(-4$ Dq)	Axial orbitals $(d_{x^2-y^2}, d_{z^2})$ rise; inter-axial orbitals (d_{xy}, d_{yz}, d_{xz}) sink. $\Delta_o = 10$ Dq.
Tetrahedral Splitting	$\Delta_t = \frac{4}{9}\Delta_o$; lower e, upper t_2	No 'g' subscript (no inversion centre). $\Delta_t < P$ always → always high-spin.
High/Low-Spin Rule	$\Delta_o < P \rightarrow$ High-Spin; $\Delta_o > P \rightarrow$ Low-Spin	Weak ligands obey Hund's rule; strong ligands force pairing into t_{2g} before e_g.
Spectrochemical Series	$I^- < Br^- < Cl^- < F^- < OH^- < ox^{2-} < H_2O < NH_3 <$ en $< CN^- < CO$	Halide donors (weak) < Oxygen donors < Nitrogen donors < Carbon donors (strong).
Fac/Mer Isomerism	Seen in octahedral $[Ma_3b_3]$	Facial = 3 identical ligands on one triangular face (90°); Meridional = in one plane.
Optical Activity	Requires non-superimposable mirror image (no plane of symmetry)	$[Co(en)_3]^{3+}$ and *cis*-$[Pt\ Cl_2(en)_2]^{2+}$ are chiral; *trans*-$[Pt\ Cl_2(en)_2]^{2+}$ is achiral (meso).
Biological Complexes	Chlorophyll (Mg^{2+}), Haemoglobin (Fe^{2+}), Vit B_{12} (Co^{3+})	Essential metal identification questions on CUET UG examination papers.

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. The coordination number and oxidation state of Cobalt in the complex $[Co(en)_2Cl_2]$ Cl are respectively:

(A) 4 and +3 (B) 6 and +3
(C) 6 and +2 (D) 4 and +2

Sol. Q1. Each bidentate ethylenediamine ("en") contributes 2 donor atoms ($2 \times 2 = 4$), plus two monodentate chlorides = coordination number 6. Oxidation state: $x + 2(0) + 2(-1) = +1 \rightarrow x = +3$. **Answer: (B)**

Q2. Which of the following compounds behaves as a double salt rather than a coordination complex when dissolved in water?

(A) $K_4[Fe(CN)_6]$ (B) $KCl \cdot MgCl_2 \cdot 6H_2O$
(C) $\left[Cu(NH_3)_4\right]SO_4$ (D) $\left[Co(NH_3)_6\right]Cl_3$

Sol. Q2. Carnallite ($KCl \cdot MgCl_2 \cdot 6H_2O$) completely dissociates in aqueous solution into simple K^+, Mg^{2+}, and Cl^- ions, behaving as a double salt. The other three retain their complex coordination entities intact. **Answer: (B)**

Q3. What is the correct systematic IUPAC name of the complex salt $K_3[Fe(CN)_6]$?

(A) Potassium hexacyanidoiron(III) (B) Potassium hexacyanidoferrate(III)
(C) Tripotassium hexacyanidoironate(II) (D) Potassium ferricyanide(III)

Sol. Q3. The cation (potassium) is named first. The complex entity is an anion, so the metal name ends in -ate: ferrate. Charge balance: $3(+1) + x + 6(-1) = 0 \rightarrow x = +3$. Name: potassium hexacyanidoferrate(III). **Answer: (B)**

Q4. The two compounds $\left[Co(NH_3)_5(NO_2)\right]Cl_2$ and $\left[Co(NH_3)_5(ONO)\right]Cl_2$ exemplify which type of isomerism?

(A) Ionisation isomerism (B) Linkage isomerism
(C) Coordination isomerism (D) Hydrate isomerism

Sol. Q4. The ambidentate nitrite ligand coordinates via nitrogen (NO_2^-, yellow) in one isomer and via oxygen (ONO^-, red) in the other, exhibiting linkage isomerism. **Answer: (B)**

Q5. A tetrahedral complex of the general formula [Mabcd] possessing four completely different monodentate ligands will exhibit:

(A) Geometrical (cis/trans) isomerism only (B) Optical isomerism (chirality) only
(C) Both geometrical and optical isomerism (D) Neither geometrical nor optical isomerism

Sol. Q5. In tetrahedral geometry, all four coordination sites are mutually adjacent (109.5°), preventing cis/trans isomerism. However, with four different ligands, the complex lacks any plane of symmetry and is optically active. **Answer: (B)**

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. In the complex $[CoF_6]^{3-}$, Cobalt(III) ($3d^6$) possesses 4 unpaired electrons. The hybridisation and orbital classification of this complex are:

(A) d^2sp^3 and inner orbital
(B) sp^3d^2 and outer orbital
(C) sp^3 and tetrahedral
(D) dsp^2 and square planar

Sol. Q6. Because F^- is a weak-field ligand, crystal field splitting is small ($\Delta_o < P$). Electrons do not pair up, forcing hybridisation to use outer 4d orbitals (sp^3d^2), creating a high-spin outer orbital complex. **Answer: (B)**

Q7. According to the Spectrochemical Series, which of the following ligands produces the STRONGEST crystal field splitting (Δ_o)?

(A) F^-
(B) H_2O
(C) NH_3
(D) CN^-

Sol. Q7. Field strength follows: $F^- < H_2O < NH_3 < CN^- < CO$. Cyanide is a carbon-donor strong-field ligand that generates a massive Δ_o, forcing low-spin configurations. **Answer: (D)**

Q8. The characteristic absorption colour of transition metal complexes in aqueous solution is fundamentally caused by:

(A) Electronic promotion between split d-orbitals ($d-d$ transitions) by absorbing visible light
(B) Emission of radiation by surrounding ligand molecules
(C) Total internal reflection across the coordination sphere
(D) Vibrational stretching of metal-ligand coordinate bonds

Sol. Q8. Visible light of wavelength $\lambda = h\frac{c}{\Delta_o}$ promotes a d-electron from lower t_{2g} to upper e_g orbitals. The transmitted optical complementary wavelength produces the perceived colour. **Answer: (A)**

Q9. The bonding in homoleptic metal carbonyls such as $[Ni(CO)_4]$ is characterized as synergic because:

(A) It involves purely ionic attraction between metal and carbon
(B) It consists exclusively of dative σ-donation from ligand to metal
(C) σ-donation from CO to metal reinforces π-back-bonding from metal to CO π^* orbitals
(D) Ligands donate electron density through four identical hydrogen bonds

Sol. Q9. σ-donation ($C \to M$) increases electron density on the metal, which enhances π-back-donation ($M \to \pi^*CO$). Each process strengthens the other, shortening and strengthening the $M-C$ bond. **Answer: (C)**

Q10. cis-Platin, the widely utilized cancer chemotherapeutic drug, corresponds to which chemical formulation?

(A) *trans*-$[Pt(NH_3)_2Cl_2]$
(B) *cis*-$[Pt(NH_3)_2Cl_2]$
(C) $[Pt(CN)_4]^{2-}$
(D) $K_2[PtCl_6]$

Sol. Q10. cis-Diamminedichloridoplatinum(II) has adjacent cis chlorides (90°) that bind to DNA guanine bases, cross-linking cancer cell DNA. The trans isomer is therapeutically ineffective. **Answer: (B)**

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. When 0.1 mol of $[Co(NH_3)_5SO_4]Br$ and 0.1 mol of $[Co(NH_3)_5Br]SO_4$ are treated separately with excess aqueous $BaCl_2$, what precipitate is obtained?

(A) Both give white precipitates of $BaSO_4$
(B) Only the second isomer gives a white precipitate of $BaSO_4$
(C) Only the first isomer gives a white precipitate of $BaSO_4$
(D) Neither isomer forms any precipitate with $BaCl_2$

Sol. Q11. Only $[Co(NH_3)_5Br]SO_4$ possesses a free sulphate counter-ion outside the coordination sphere to react with Ba^{2+}, precipitating white $BaSO_4$. The sulphate in the first isomer is non-ionisable. **Answer: (B)**

Q12. Which of the following octahedral complexes exhibits optical isomerism (exists as non-superimposable *d* and *l* enantiomers)?

(A) *trans*-$[Co(en)_2Cl_2]^+$
(B) *cis*-$[Co(en)_2Cl_2]^+$
(C) $[Co(NH_3)_4Cl_2]$
(D) $[Cr(NH_3)_6]^{3+}$

Sol. Q12. *cis*-$[Co(en)_2Cl_2]^+$ lacks a plane of symmetry and is chiral (optically active). In contrast, the trans isomer possesses an internal horizontal plane of symmetry, rendering it achiral and optically inactive. **Answer: (B)**

Q13. In a tetrahedral crystal field, the ratio of crystal field splitting Δ_t to octahedral splitting Δ_o for the same metal and ligand set is approximately:

(A) $\Delta_t = \frac{9}{4}\Delta_o$
(B) $\Delta_t = \frac{4}{9}\Delta_o$
(C) $\Delta_t = \frac{1}{2}\Delta_o$
(D) $\Delta_t = \frac{2}{3}\Delta_o$

Sol. Q13. Because there are only 4 ligands (not 6) and ligands point between orbital axes, the tetrahedral crystal field splitting energy is small: $\Delta_t = \frac{4}{9}\Delta_o$. Consequently, tetrahedral complexes are always high-spin. **Answer: (B)**

Q14. What are the respective metal centers present in Chlorophyll, Haemoglobin, and Vitamin B_{12}?

(A) Fe, Mg, Co
(B) Mg, Fe, Co
(C) Co, Fe, Mg
(D) Cu, Fe, Zn

Sol. Q14. Chlorophyll is a coordination complex of Magnesium (Mg^{2+}), Haemoglobin contains Iron (Fe^{2+}), and Vitamin B_{12} (cyanocobalamin) contains Cobalt (Co^{3+}). **Answer: (B)**

Q15. Which of the following complex ions has the highest thermodynamic stability constant (β) in aqueous solution?

(A) $[Fe(H_2O)_6]^{3+}$
(B) $[Fe(NH_3)_6]^{3+}$
(C) $[Fe(ox)_3]^{3-}$
(D) $[FeCl_6]^{3-}$

Sol. Q15. $[Fe(ox)_3]^{3-}$ contains three bidentate oxalate ligands that form three 5-membered chelate rings around iron. The chelate effect substantially increases thermodynamic stability ($\Delta S > 0$) relative to monodentate complexes. **Answer: (C)**

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